



Dept. of Computer Science and Engineering

Jashore University of Science and Technology

Faculty of Engineering and Technology - FET

Lab Report : 03

ER Diagram and Database structure

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Course Details:

**Course Title: Database Management
System Lab**

Course Code: CSE 2204

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Project Name: WellBeing – A Mental Health Improver Platform

Project Description:

WellBeing is an interactive web platform crafted to empower individuals on their mental health*
Comprehensive Mood Tracking: Users log daily moods via an intuitive emoji slider or dropdown (ranging from Very Happy to Very Sad), optionally adding contextual notes or journal reflections.

* Insightful Analytics Dashboard: Visualize mood trends over configurable timeframes using interactive charts, view weekly average scores, and monitor consistency streaks to foster self-awareness.

* Tailored WellBeing Tips: Leverage a curated library of self-care suggestions—daily affirmations, breathing exercises, and mindfulness prompts—dynamically recommended based on recent emotional data.

* Private Journaling Module: A secure diary feature where users can record thoughts, tag entries, and search or filter past reflections for personal review.

* Therapeutic Resource Hub : Browse expert profiles complete with specialties, availability, and contact options. Access emergency hotlines and community support links.

* Appointment Scheduling: Seamless booking interface to request consultations, with status tracking—pending, confirmed, or cancelled—and calendar integration.

Under the hood, a relational database ensures data integrity, while session management and encryption uphold user privacy. Responsive layouts guarantee accessibility across devices, with emphasis on calming UI and user-centric navigation.

Functional Requirements:

Secure Authentication: Signup and login flows with encrypted credentials, session handling, and validation.

Responsive Home Interface: Clean landing page with guided CTAs and adaptive navigation menu.

Mood Entry Workflow: Interactive form for mood submission, storing timestamped entries in the database.

Analytics Dashboard: Endpoint-driven chart rendering for mood history, averages, and streak indicators.

Tips & Activities Section: Dynamic suggestion listing and optional activity logging for self-care tasks.

Journal Management: Create and retrieve rich-text diary entries, with search and preview functionalities.

Resources Directory: Display profiles for mental health professionals and vital support links.

Booking Module: Submit and manage appointment requests with relational linking between users and professionals.

Database Schema Details:

This database schema represents a Mental Health Management System, designed to support user journaling, mood tracking, therapy appointments, and personalized suggestions. Here's a concise summary:

Users Table: Stores personal user details including name, email, age, gender, and account creation date.

Journals Table: Allows users to write journal entries with associated timestamps for emotional tracking.

Mood Entries Table: Users log their moods and mood scores along with optional notes and timestamps.

Suggestions Table: Provides mental health tips or advice based on mood entries.

Therapists Table: Contains therapist details such as name, specialization, email, and availability.

Appointments Table: Connects users with therapists, tracking appointment dates and statuses.

Relationships:

Users can have multiple journal entries, mood logs, and appointments.

Each appointment links a user to a therapist.

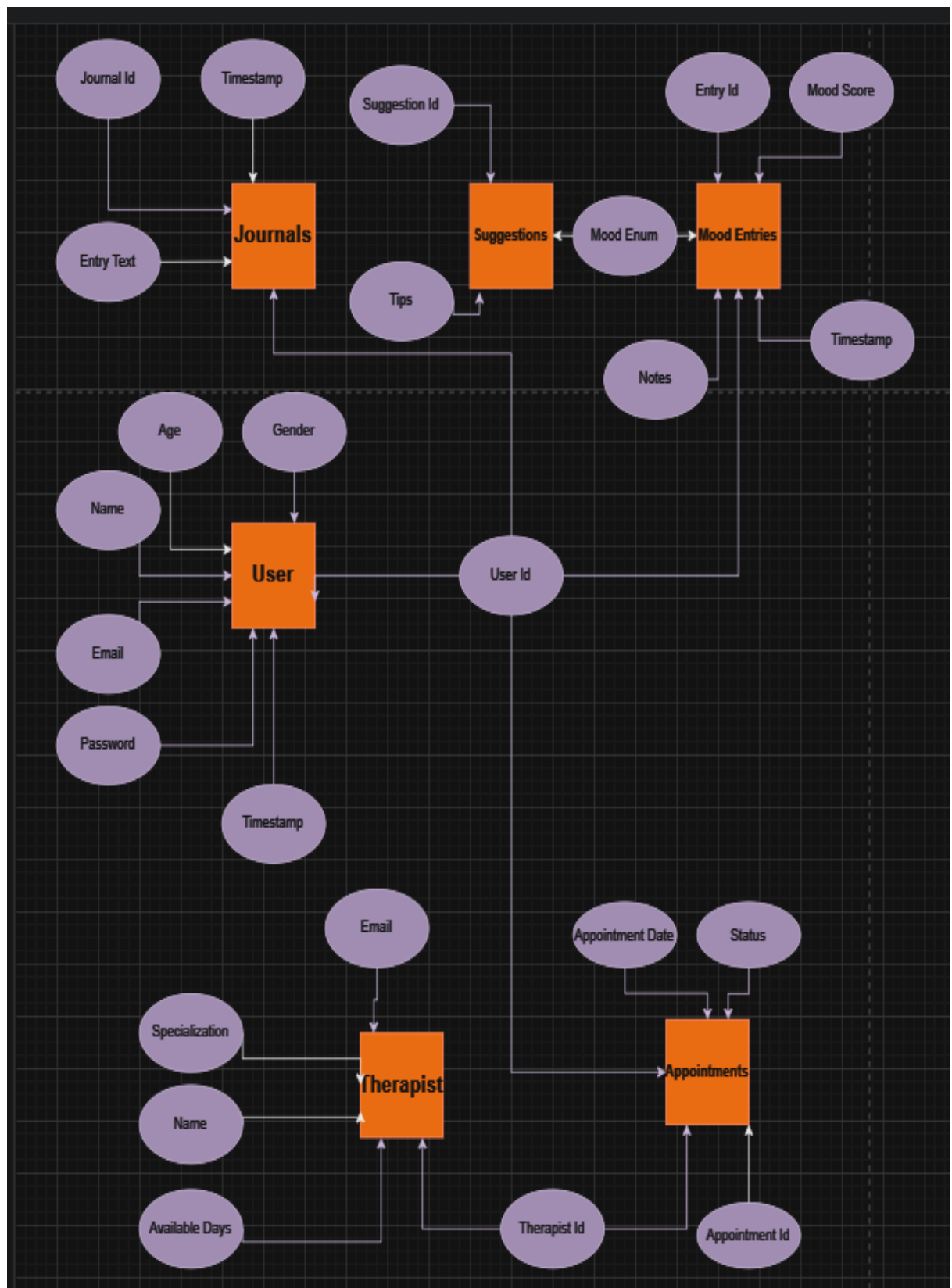
Suggestions may be generated in response to mood entries.

Enums: Gender, mood types, and appointment statuses are standardized using ENUM fields.

Data Types: Includes VARCHAR for text, INT for identifiers and scores, TEXT for longer inputs, and timestamps for time-based tracking.

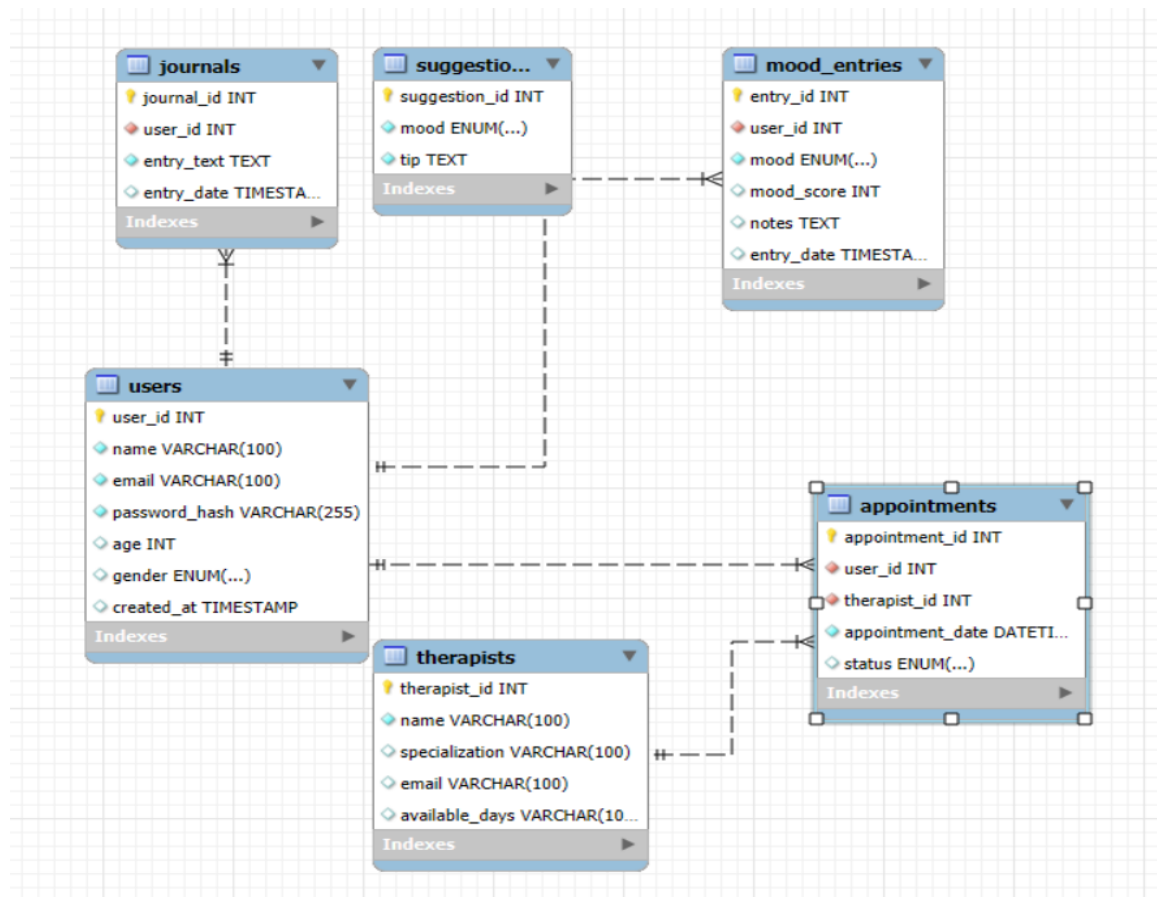
Use Case: Ideal for supporting mental well-being by combining self-reflection, emotional monitoring, professional therapy, and tailored advice.

Database Design:
Figure 1. EER Diagram



Database Design (MySQL)

Figure 2. EER Diagram



SQL Schema Sample

```
CREATE DATABASE IF NOT EXISTS wellbeingjust;
USE wellbeingjust;

-- 1. Users table
CREATE TABLE users (
  user_id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  age INT,
  gender ENUM('Male', 'Female', 'Other') DEFAULT 'Other',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. Mood Entries
CREATE TABLE mood_entries (
  entry_id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  mood ENUM('Very Happy', 'Happy', 'Neutral', 'Sad', 'Very Sad') NOT NULL,
  mood_score INT CHECK (mood_score BETWEEN 1 AND 5),
  notes TEXT,
  entry_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
);

-- 3. Journals
CREATE TABLE journals (
  journal_id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  entry_text TEXT NOT NULL,
  entry_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
);
```

-- 4. Suggestions (Tips)

```
CREATE TABLE suggestions (  
  suggestion_id INT AUTO_INCREMENT PRIMARY KEY,  
  mood ENUM('Very Happy', 'Happy', 'Neutral', 'Sad', 'Very Sad') NOT NULL,  
  tip TEXT NOT NULL  
);
```

-- 5. Therapists

```
CREATE TABLE therapists (  
  therapist_id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL,  
  specialization VARCHAR(100),  
  email VARCHAR(100),  
  available_days VARCHAR(100)  
);
```

-- 6. Appointments

```
CREATE TABLE appointments (  
  appointment_id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT NOT NULL,  
  therapist_id INT NOT NULL,  
  appointment_date DATETIME NOT NULL,  
  status ENUM('Pending', 'Confirmed', 'Cancelled') DEFAULT 'Pending',  
  FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE,  
  FOREIGN KEY (therapist_id) REFERENCES therapists(therapist_id) ON DELETE CASCADE  
);
```